LUMEXA
Space-themed coding education for K–12

AGES 12–18 | BEGINNER FRIENDLY

CURRICULUM OVERVIEW

Web Developer Path

Build 3 Live Websites

A complete, project-driven journey from a first HTML page to a deployed, full-stack React application. Students don't just learn about the web — they write the code, style real interfaces, and ship three live, working websites anyone can visit.



3 LIVE WEBSITES TO LAUNCH
Real addresses anyone can visit

4 TESTS ALONG THE WAY
3 course tests + 1 final test

3	24	28	12–18
Courses	Lessons	Sessions in total	Age range

WHO IT IS FOR
Beginner friendly — start with your first web page

BEFORE YOU START
No coding experience needed

LEARNING PROGRESSION

01 HTML and CSS Mastery
Structure and style responsive websites, then go live.

02 JavaScript and Interactivity
Pages that react to clicks, fetch live data and save it.

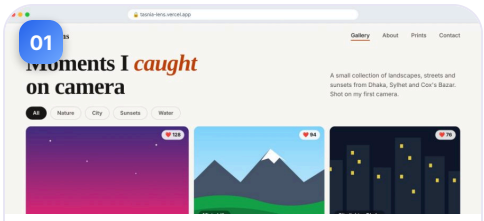
03 React and Full-Stack Web
Modern React apps, deployed live like a real team would.

Learn by shipping, not by watching
Every lesson pairs a real idea with running code that deploys live on the web.

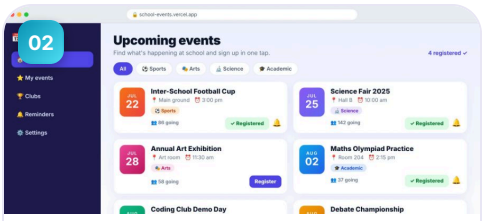
A portfolio, not just a certificate
Students finish with three live websites they can show, share and keep building on.

Tested, so progress is real
Every course ends with a test, and a final test closes the program, so progress is proven, not assumed.

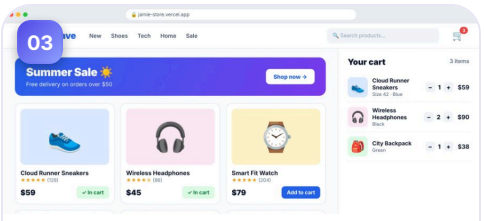
HANDS-ON PROJECTS STUDENTS BUILD



Photography Portfolio
HTML · CSS · JavaScript



School Events App
HTML · CSS · JavaScript



E-Commerce Store
React · Node.js

COURSE 01 · LESSONS 1–8

01 HTML and CSS Mastery

Build beautiful, responsive websites that look great on any device

HTML5 · CSS3 · Flexbox · CSS Grid · Vercel

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

Your child learns the fundamentals of the web — HTML structure and CSS styling — and builds their first real, deployed website that anyone can visit.

PORTFOLIO PROJECTS

- Personal landing page
- Responsive portfolio layout
- Product showcase site

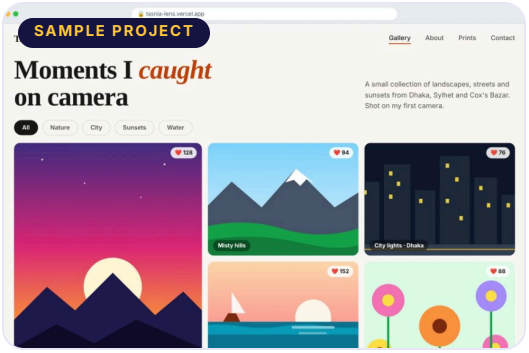
WHY IT MATTERS

HTML and CSS are the base layer of every website on the internet. Students learn to structure content and design layouts that adapt to any screen — the skill every later project builds on.

LESSON BREAKDOWN

1. How the web works (browsers, servers, HTML)
2. HTML structure and semantic elements
3. CSS basics: colors, fonts, spacing
4. Flexbox for modern layouts
5. CSS Grid for complex designs
6. Responsive design and mobile-first
7. Animations and hover effects
8. Deploying to Vercel and sharing

8 LESSONS — TEST UNLOCKS



HANDS-ON PROJECT

Photography Portfolio

HTML · CSS · JavaScript

A responsive photo-gallery website, published live — structure, style and layout that adapt to any screen.

COURSE TEST

HTML and CSS Mastery — Course Test

Checks what you built and learned in HTML and CSS Mastery. Taken after lesson 8, before the next course begins.

30 multiple-choice · 10 true / false · 40 marks

COURSE 02 · LESSONS 9–16

02 JavaScript and Interactivity

Add dynamic, interactive behavior to any webpage

JavaScript (ES6+) | DOM API | Fetch API | Local Storage

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

Static websites are just the beginning. Your child learns JavaScript to make pages that react to clicks, fetch live data, and behave like real apps.

PORTFOLIO PROJECTS

- Interactive to-do app
- Live weather dashboard
- Memory card game

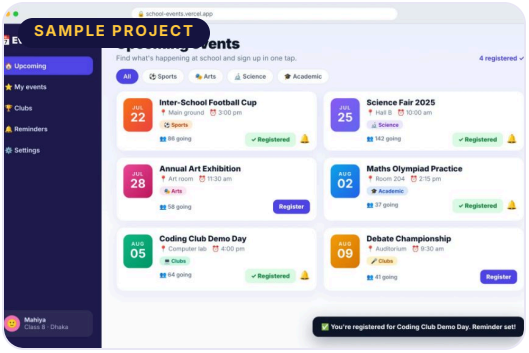
WHY IT MATTERS

JavaScript is what turns a static page into a real application. Students learn to handle user actions, talk to live APIs, and save data — the core skills behind every interactive website.

LESSON BREAKDOWN

1. JavaScript fundamentals (variables, functions, loops)
2. DOM manipulation: selecting and changing elements
3. Event listeners: responding to user actions
4. Fetch API: pulling live data from the web
5. Async/await and working with APIs
6. Local Storage for saving user data
7. Building a complete interactive app
8. Debugging and polishing

8 LESSONS — TEST UNLOCKS



HANDS-ON PROJECT

School Events App

HTML · CSS · JavaScript

An events app that filters, registers and remembers — a page that reacts to clicks and keeps its data.

COURSE TEST

JavaScript and Interactivity — Course Test

Checks what you built and learned in JavaScript and Interactivity. Taken after lesson 8, before the next course begins.

30 multiple-choice | 10 true / false | 40 marks

COURSE 03 · LESSONS 17–24

03 React and Full-Stack Web

Build and deploy modern React applications used in real companies

React 18 | Next.js | Vercel | REST APIs

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

React is the most in-demand frontend framework in the world. Your child builds reusable components, manages state, fetches data from APIs, and deploys a full React app live.

PORTFOLIO PROJECTS

- E-commerce product page
- Social media feed clone
- Full portfolio React app

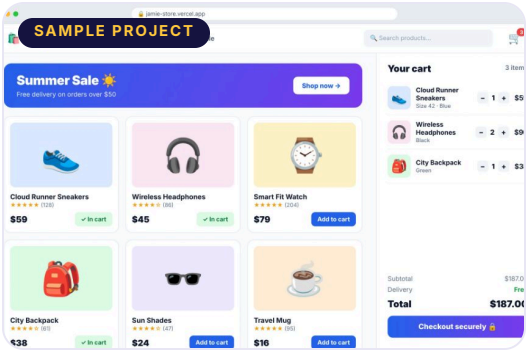
WHY IT MATTERS

React and Next.js power the majority of modern production websites. Students learn the same component model, data-fetching patterns, and deployment workflow used by real engineering teams.

LESSON BREAKDOWN

1. Why React exists and how components work
2. JSX, props, and component trees
3. State with useState and event handling
4. Fetching data with useEffect
5. React Router and multi-page apps
6. Next.js and server-side rendering
7. Styling with Tailwind CSS
8. Deploying to Vercel and final project

8 LESSONS — TEST UNLOCKS



HANDS-ON PROJECT

E-Commerce Store

React · Node.js

An online shop with product pages, a working cart and a checkout flow, built from reusable React components.

COURSE TEST

React and Full-Stack Web — Course Test

Checks what you built and learned in React and Full-Stack Web. Taken after lesson 8, before the next course begins.

30 multiple-choice | 10 true / false | 40 marks

HOW STUDENTS ARE TESTED · 4 TESTS

Level-Up Checkpoints

A website that looks right can still be built wrong. Course tests check the ideas under the pixels — HTML structure, CSS layout, JavaScript behaviour and React components — and a final test closes the path.



THE CHECKPOINT PATH — EVERY COURSE ENDS WITH A TEST, THEN ONE FINAL



COURSE TEST · after every 8 lessons

Show what you built and learned

After every 8 lessons a course test checks the ideas and skills of that course, so each next course starts on solid ground.

- **30** multiple-choice questions — choose one of four answers
- **10** true / false questions
- **40** marks in total — 1 mark for every question

FINAL TEST · the finish line

One final test on the whole curriculum

One final test across HTML & CSS, JavaScript and React: questions plus a small practical project.

- **25** multiple-choice questions
- **5** true / false questions
- **20** marks for one small practical project (about 15 minutes) that your teacher marks

A TASTE OF A TEST

SAMPLE · MULTIPLE-CHOICE

Which CSS feature lays items out in a flexible row or column?

- A Flexbox ✓ CORRECT
- B Alt text
- C A favicon
- D A hyperlink

SAMPLE · TRUE / FALSE

JavaScript can change what is on a page after it has loaded.

True ✓ | False

Examples in the style of a real test — written for this document, not taken from one.

AFTER EVERY TEST

Saved as you go
Answers save automatically while a student works, so nothing is lost if the connection drops.

Checked by the teacher
The teacher reviews the report and shares the result with the student.

Fair for everyone
Questions and answer options appear in a different order for every student.

The Complete Journey

Across 3 courses, 24 lessons and 4 tests, students progress from writing their first HTML page to building and deploying a full-stack React application — walking away with three real, live websites for their portfolio.

3 COURSES · 24 LESSONS

01 HTML and CSS Mastery
Personal landing page · Responsive portfolio layout · Product showcase site
8 lessons · + course test

02 JavaScript and Interactivity
Interactive to-do app · Live weather dashboard · Memory card game
8 lessons · + course test

03 React and Full-Stack Web
E-commerce product page · Social media feed clone · Full portfolio React app
8 lessons · + course test

HOW STUDENTS ARE TESTED · 4 TESTS

COURSE TEST 1
HTML and CSS Mastery

COURSE TEST 2
JavaScript and Interactivity

COURSE TEST 3
React and Full-Stack Web

FINAL TEST
One final test on the whole curriculum
One final test across HTML & CSS, JavaScript and React: questions plus a small practical project.

BY THE END, STUDENTS CAN

- Structure and style a fully responsive website from scratch using HTML and CSS.
- Add real interactivity with JavaScript, including live data from external APIs.
- Build and deploy a modern, full-stack React application live on the web.

Ready to start building?
All lessons and projects are taught live by an expert Lumexa instructor.

Lumexa Web Developer · Ages 12–18